

Project Documentation-Team Py07

Project Title:

Password Manager Application

Project Overview:

Objective: To provide a secure, efficient, and user-friendly tool for managing passwords which aims to ensure that users can store, retrieve, and manage their passwords with ease while maintaining a high level of security through state-of-the-art encryption and authentication mechanisms.

The **Scope** of this project includes the development, deployment, and maintenance of the password manager application that incorporates various security features and functionalities essential for protecting sensitive information.

Project Team:

1. Shreyjith S:
Cross-Platform Compatibility, Testing and Quality Assurance, User Documentation and Support.
2. Shrikant Mhaske:
User Interface, Authentication and Authorization, Password Encryption and Security, Password Generation.
3. Nikita Bansod:
Password Storage, Account Management, Password Organization.
4. Mohd Tabrej:
Search and Retrieval and Backup and Synchronization.
5. Kavita Malepati:
Privacy and Data Protection and Security Best Practices.

Project Technologies:

- **Graphical User Interface (GUI):** It uses tkinter for creating the GUI elements such as windows, buttons, labels, and dialogs
- **Encryption:** It uses cryptography.fernet.Fernet for encrypting and decrypting data, ensuring that stored passwords are secure.
- **Password Hashing:** It uses bcrypt for hashing passwords, adding an extra layer of security by storing only hashed versions of passwords.
- **Database Management:** It uses sqlite3 for managing a local SQLite

database, which stores user data and passwords.

- **Email Sending:** It uses smtplib for sending emails, including OTPs for verification and greeting emails to users.
- **Random and Secure Password Generation:** It uses random, string, and secrets libraries for generating random passwords and OTPs.
- **PrettyTable:** It uses prettytable for formatting and displaying data in a tabular form.

Project Setup:

To install the Password Manager, follow these steps:

- Ensure you have Python installed on your system. You can download it from python.org.
- Install the following libraries: tkinter, bcrypt, sqlite3, datetime, cryptography, smtplib, random, string, secrets, prettytable and argparse, maskpass, sys.
- Download the `Final\_Password\_Manager.py` and `Final\_UI.py` script to your preferred directory. Run `Final\_UI.py`.

Project Resources/ Reference:

- Libraries Documentation
- Youtube
- StackOverflow
- GeekForGeeks
- W3Schools SQL Tutorial

Project Risks:

- **Security Breach:** Unauthorized access to the password database through hacking or exploitation of vulnerabilities.

Mitigation Strategies:

- Implement strong encryption algorithms (e.g., AES) for all stored passwords.
- Conduct regular security audits and penetration testing.
- Apply timely patches and updates to fix any discovered vulnerabilities

- **Data Loss:** Accidental deletion or corruption of password data.

Mitigation Strategies:

- Perform regular backups of the database.
- Implement robust recovery and restore mechanisms.
- Conduct thorough testing of all database operations to ensure reliability
- **System Downtime:** Unexpected system downtime due to server issues or software bugs.

Mitigation Strategies:

- Implement robust server infrastructure with redundancy and failover mechanisms.
- Conduct regular maintenance and monitoring of servers.
- Use automated monitoring and alert systems to detect and respond to issues promptly

Additional Comments:

- Innovation and Unique Selling Points:
 - The integration of two-factor authentication (2FA) for added security sets the Password Manager apart from many basic password managers.
 - The built-in password generator with customizable options (length, complexity, inclusion of special characters) provides users with strong, unique passwords, enhancing overall security.
- User Education and Support:
 - The user documentation is comprehensive, covering installation, setup, usage, troubleshooting, and FAQs, ensuring users can easily understand

and use the application.

- Continuous user education on best practices for password management is crucial for maintaining high security standards.

---THANK YOU---

**By Team Py07,
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